

AI Assignment

Deadline – Sunday EOD

Multi-Agent RAG System with FastMCP Integration

Assignment

Build a **multi-agent AI system** using **FastAPI** that answers FAQ queries using **Retrieval-Augmented Generation (RAG)**, provides **weather information**, and manages tasks through a **FastMCP Todo server**.

The system must include a **parent agent (orchestrator)** that routes requests to two specialized agents.

Multi-Agent Framework

You may implement the multi-agent system using:

- **Semantic Kernel (preferred)**
- LangGraph
- LangChain
- Any other suitable framework

Agents

Parent Agent (Orchestrator)

Responsible for coordinating the workflow.

Responsibilities:

- Receive user queries from the API
- Detect user intent
- Route requests to the appropriate agent
- Return the final response

RAG Agent

Handles FAQ queries using a **RAG pipeline**.

Responsibilities:

- Load FAQ data from **CSV files**
- Build a **vector database from the CSV dataset**
- Generate embeddings and perform semantic retrieval
- Generate responses only from retrieved context
- Return a fallback message if the answer is not found

Tool Agent

Handles external tools.

Supported tools:

Weather Tool

Returns the current weather for the configured city.

FastMCP Todo Server

Integrate a Todo service using **FastMCP** supporting:

- Create task
- List tasks
- Update task
- Delete task

APIs

Chat API

POST /chat

Accepts a query and routes it through the parent agent.

Weather API

GET /weather

Returns current weather for the configured city.

Todo APIs (*Optional*)

Expose REST endpoints for creating, listing, updating, and deleting tasks that interact with the **FastMCP Todo server**.

Authentication

All APIs must use **JWT authentication**.

Testing

Implement **happy-path tests** for the main APIs using a framework such as **Pytest**.

Acceptance Criteria

- Parent agent orchestrates request routing
- RAG agent retrieves answers from a vector database built from CSV data
- Tool agent successfully invokes the weather tool and FastMCP Todo server
- /chat and /weather APIs function correctly
- JWT authentication protects APIs
- Happy-path tests run successfully
- Project includes setup instructions in Readme and runnable code
- Must submit GitHub repo

Good to have

- Docker setup for running the service.

Kindly use ONLY mini models - 5 mini OR 4.1 mini.

Open API key

sk-proj-

folxYVuoejYLNh5AWG3RKLRfKJI_UR_eAZo_rF1iDxGjgPLNyHft00BOBQzKgN0GhP18GdeT3BlbkFJcffL_zZVx
UzECTSGnYJ_FAoZMe5iFI_573ND-Mf5OHju_YqdIYLy6vnnvE4QQnDdmQ83SUpR0kA